

MKTG 440 – Marketing Research
Spring 2025
Exam 1
Version A

Printing and signing your name	5 points
Multiple-choice	60 points
Short answer	35 points
	<u>100 points</u>

You will have 70 minutes to complete this exam.

Allowed resources: You must complete this exam on your own. You may use one letter-size page of notes (front and back). No other materials may be used.

Academic integrity: You may not discuss, share, record, or post any exam content or your answers with anyone.

Clarifying questions: Please do not ask questions during the exam. The content and wording of each question has been carefully reviewed. However, if you think a question is ambiguous and can only be answered by making an additional assumption, you may use Short Answer Question 4 to explain your reasoning and provide your answer.

When you finish the exam: Return your answer sheet, this entire exam, and your notes page.

Please sign your name below to indicate that you:

- have read and understood the instructions for this exam
- will submit answers that are based solely on your own work
- will not share either the questions or the answers to this exam with anyone

Print your name

Signature

MULTIPLE-CHOICE [20 questions, 3 points each].

Please mark your answer to each multiple-choice question on your bubble sheet. Only bubble sheet answers will be graded. You will not lose points for the first multiple choice question you answer incorrectly.

1. Classify the following questions into management or research:

- I. *"What streaming content do consumers value?"*
- II. *"Which consumer segments would switch to a competitor given a \$5 price increase?"*
- III. *"How can we increase subscriber growth?"*

- A. I-management, II-research, III-management
- B. I-research, II-management, III-research
- C. I-management, II-research, III-research
- D. I-research, II-research, III-management
- E. I-research, II-management, III-management

2. A key advantage of *stratified sampling compared to simple random sampling* is:

- A. It is quicker to conduct.
- B. It requires no prior knowledge of population characteristics.
- C. It ensures key subgroups are represented in the sample.
- D. It eliminates sampling error.

3. A marketer shows two versions of an online advertisement to different consumers to see which version leads to higher click-through rates. This is an example of:

- A. Descriptive research
- B. Observational research
- C. Causal research
- D. Focus group research
- E. Randomized research

4. Which of the following increases *internal validity* in an experiment?
- A. Increasing sample size
 - B. Randomly assigning participants to groups
 - C. Conducting the study in a natural environment
 - D. Using a convenience sample
5. Which of the following statements about the marketing research process is **INCORRECT**?
- A. Before choosing a research method and collecting data, we should define the research questions we are interested in answering.
 - B. Managers, unlike researchers, have a better grasp of the underlying root problems that contribute to business decision problems.
 - C. The structure of the marketing research process is useful because people tend to be biased in how they process complex information.
 - D. The marketing research process is iterative (it repeats itself) because the marketing environment is frequently changing.
6. Which of the following is an advantage of *qualitative research methods* compared to *quantitative research methods*?
- A. Qualitative research is better suited to understanding causal relationships.
 - B. Qualitative research is better suited to generating ideas and clarifying concepts.
 - C. Qualitative research uses large samples and more structure, and is therefore more generalizable.
 - D. Qualitative research can use secondary data sources like store audits and consumer panels.
7. What type of research is best for understanding customer motivations and feelings?
- A. Quantitative research
 - B. Causal research
 - C. Qualitative research
 - D. Descriptive research
 - E. Observational research

8. The “Iceberg Principle” in marketing research suggests that:
- A. Research is often expensive and should be minimized.
 - B. Most marketing problems are visible to decision-makers.
 - C. Underlying causes are hidden beneath surface-level symptoms.
 - D. Trends in consumer behavior are slow-moving and predictable.
9. Which of the following best describes a scenario where using a projective technique would be more effective than direct questioning?
- A. Asking about a topic that is concrete, but for which consumers struggle with accurate recall.
 - B. Exploring subconscious feelings about a controversial brand that participants may not openly express.
 - C. Determining the exact price point at which customers would stop purchasing a product.
 - D. Measuring customer satisfaction with a new product by asking consumers to rate it on multiple dimensions.
10. Which of the following is an example of a “discovery-oriented” (descriptive) research objective rather than a “strategy-oriented” (prescriptive) one?
- A. Determine which pricing plan will best reduce churn.
 - B. Identify the top five reasons our customers are dissatisfied.
 - C. Choose the most effective promotional campaign among three alternatives.
 - D. Decide whether to invest in a new product launch.

11. Consider the following survey question.

What is your race/ethnicity? Mark all that apply.

- ☐ *Asian/Pacific Islander*
- ☐ *Black/African American*
- ☐ *Caucasian/White*
- ☐ *Hispanic/Latinx*
- ☐ *Middle Eastern/North African*
- ☐ *Native American/Alaska Native*
- ☐ *An identity not listed*

How many possible ways are there for a survey respondent to answer the above question?

- A. 1
- B. 7
- C. $7^2 = 49$
- D. $2^7 = 128$

12. Which of the following scales measuring academic performance have the **RATIO** property?

- I. Letter grade
- II. Number of questions answered correctly on an exam
- III. Percent (%) of questions answered correctly on an exam

- A. I only
- B. II only
- C. III only
- D. II and III only
- E. I, II, and III

13. Chris is an HR Manager at Arizona State University and is interested in studying whether allowing pets in classrooms can improve student morale and decrease stress. Chris randomly selects one course from the spring semester course catalog in which to run an experiment. He convinces the course instructor to allow students to bring their pets to class for two weeks. At the end of the two weeks, he gives all students in the class a short survey to measure their stress levels. Chris finds that the students who brought pets into class, which he calls the treatment group, had lower stress levels than the students who did not bring pets into class, which he calls the control group. What, if anything, is wrong with Chris's experimental setup?

- A. "Correlation/association" criterion is not met
- B. "Temporal antecedence" criterion is not met
- C. "No third factor" criterion is not met
- D. All three criteria are met. (Nothing is wrong)

14. Which of the following is a way to eliminate *sampling error* when data are collected via a questionnaire?

- I. Sample every individual in the target population.
- II. Provide incentives for participating to reduce non-response bias.
- III. Avoid asking sensitive questions in the questionnaire.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

15. Which of the following is **TRUE** about cluster sampling?

- A. By sampling from a subset of clusters, we can reduce costs and increase efficiency.
- B. It works best when elements in the same cluster are very similar, but elements in different clusters are heterogeneous.
- C. We need a sampling frame for every cluster in the population.
- D. Cluster sampling allows us to obtain a more representative sample than other probability sampling methods.

16. Which statement about the block of survey questions below is **INCORRECT**?

*To what extent do you agree with the following statements?
ChatGPT is...*

	<i>Strongly agree</i>	<i>Agree</i>	<i>Neither agree nor disagree</i>	<i>Disagree</i>	<i>Strongly disagree</i>
<i>1. User-friendly</i>	()	()	()	()	()
<i>2. My go-to source for doing research</i>	()	()	()	()	()
<i>3. Highly accurate</i>	()	()	()	()	()
<i>4. A tool that makes society better off</i>	()	()	()	()	()

- A. This series of questions likely reflects an attempt to measure an abstract construct such as consumer sentiment towards ChatGPT.
- B. We can add or average responses across these four questions to calculate a score that reflects attitude towards ChatGPT or a related construct.
- C. These questions use a Likert Scale, which measures strength of belief using a nominal scale.
- D. To add or average responses, we must first assign numbers to each scale level and assume that the distance between scale levels is equal.

17. Which of the following question types is **LEAST** likely to suffer from *response error*?

- A. Do you think the product is affordable and high quality?
- B. Don't you agree that the product is the best on the market?
- C. Would you pay \$9.95 to use the product for 12 months?
- D. How many times did you use the product in the last 12 months?

18. Suppose you are interested in creating a sample of 1,000 consumers from the whole US population. You want to make sure that your sample has individuals from different income levels, and you are particularly interested in ultra-high net-worth individuals, defined as individuals with assets exceeding \$10 million. Your concern is that these individuals represent a very small fraction of the population, less than 1%. Which of the following survey methods would be **LEAST** appropriate to ensure your sample contains a sufficient number of ultra-high net-worth individuals?
- A. Systematic sampling
 - B. Stratified sampling
 - C. Quota sampling
 - D. Judgmental sampling
19. Which of the following options is **NOT** an example of *non-sampling error*?
- A. Your sample is representative of the population, but the sample average income is not equal to the population average income.
 - B. Half of the individuals in your sample received your survey link but never opened it.
 - C. Five of the individuals in your sample were included in your sampling frame, but aren't part of the target population you are interested in.
 - D. Your sampling frame is based on an outdated list, and therefore no members of your target population less than 25 years of age are included in your sample.

20. A beverage company wants to evaluate consumer satisfaction with its new energy drink, WiredAF. The company includes the following question in a survey sent to customers who have previously bought the drink:

How epic is WiredAF? Choose the option below that best represents how much you love it!

- *It's bangin'!*
- *Turbo love it*
- *Really love it*
- *Kinda love it*
- *Sorta love it*

Which of the following are **TRUE** observations about this question design?

- I. The question is a leading question.
- II. The responses are not measured using an ordinal scale.
- III. We cannot use mode to measure central tendency in the responses.

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II, and III

SA1 [10 points]. GreenWave Inc. is launching a new line of eco-friendly cleaning products. The company wants to understand consumers' attitudes towards and purchasing behavior related to sustainable products. You are tasked with designing part of their customer survey.

A. **[2 points]** Which of the following questions should come first in the survey? Why?

Question 1: How important is sustainability to you when choosing cleaning products?

Question 2: What factors are most important to you when choosing cleaning products?

B. **[2 points]** Consider the following survey question:

Are you satisfied with the price and effectiveness of GreenWave's cleaning products?

○ Yes

○ No

What, if anything, causes this *question* to be problematic? Rewrite it to resolve the issue.

C. **[2 points]** Consider the following survey question:

Why do you use harmful chemical-based cleaning products when eco-friendly options like GreenWave's are available?

What, if anything, causes this *question* to be problematic? Rewrite it to resolve the issue.

D. **[2 points]** Propose an ordinal scale question that can be used to measure how much consumers budget for purchasing cleaning products.

E. **[2 points]** GreenWave intends to distribute this survey to a database of its regular customers. How might the results to the question you proposed above differ if the survey is sent only to GreenWave's regular customers as opposed to if it is sent to a random sample of people in the US population?

SA2 [10 points]. A checkout display is a strategically placed in-store advertising fixture found at the checkout area. It is designed to capture the attention of customers as they complete their purchases at the end of their shopping trip, often promoting impulse-buy products such as snacks, small gadgets, or magazines.

Javier is a manager at Wegmans, a Northeast US supermarket chain, and is conducting an experiment to investigate whether including candy bars in store checkout displays causes shoppers to increase the total amount they spend on groceries. He randomly selects 14 days in 2024 to run an experiment. On 7 of those days, he fills the checkout displays with candy bars. On the other 7 days, he fills the checkout displays with chewing gum. To measure the effect of the display contents on grocery spending, he adds up the total amount spent on *all grocery items except for candy bars and chewing gum* for each receipt over the 14-day period. Javier finds that shoppers who purchased items from the candy bar display spent less on other groceries than shoppers who purchased items from the chewing gum checkout display.

A. **[2 points]** Does Javier's experiment satisfy the "Correlation/association" criterion? Why or why not?

B. **[2 points]** Does Javier's experiment satisfy the "Temporal antecedence" criterion? Why or why not?

C. **[2 points]** Does Javier's experiment satisfy the "No third factor" criterion? Why or why not?

D. **[2 points]** Does Javier's experiment have high external validity? Why or why not?

E. **[2 points]** Does Javier's experiment have high internal validity? Why or why not?

SA3 [10 points]. Campus Health administers the Health and Wellness Survey each year to collect information about University of Arizona students' health-related behaviors, including substance use, mental health, sexual activity, nutrition, etc.

- A. **[2 points]** What might Campus Health use as a sampling frame? Describe any potential sources of sampling frame error with your proposed approach.
- B. **[2 points]** Suppose Campus Health wants to ensure that their survey is administered to an equal number of freshmen, sophomores, juniors, and seniors. Describe a probability sampling approach that they could use to achieve this objective.
- C. **[2 points]** Now imagine that Campus Health needs to administer their survey in person. Due to staffing limitations, they decide to visit only 25% of classes in the Spring semester for the purposes of administering their survey. They administer the survey to every student in the courses that they visit. What type of probability sampling approach does this describe? Under what conditions will it result in a representative sample?

- D. **[2 points]** Campus Health is considering including the following question in next year's Health and Wellness Survey. Identify one problem with the question and propose a modification to improve it.

How many kilograms of marijuana have you smoked in the last month?

- E. **[2 points]** The Health and Wellness Survey includes the following text in the intro of the survey.

You are invited to voluntarily complete this survey. The Campus Health Service Health and Wellness Survey is administered annually to collect information about University of Arizona students' health-related behaviors including substance use, mental health, sexual activity, nutrition, etc. There are no guaranteed benefits associated with completing this survey; but the information will be used to develop programming and to analyze health trends. The aggregate data may also be included in publications, presentations and reports. **This survey takes approximately 15 minutes to complete.** You will not be compensated for your participation. Please read each item carefully.

This survey is anonymous, so please answer honestly.

Provide a specific example from the text above that may decrease non-response bias.

SA4 [OPTIONAL]. If you feel a question's wording is ambiguous, you may write a brief explanation of your interpretation and proceed based on that interpretation. Clearly denote the question number and write your response in the following format.

1. Question number
2. Your answer
3. Your explanation

Extra Page 1

Extra Page 2